

Bufan Gao

github.com/jouisseuse · bufan@uchicago.edu

1. Research Interests

Multi-agent Systems in Humans and AI; Human cognitive intelligence; Computational Psychology

2. Education

The University of Chicago	2025 — Present
PhD Student in Social Psychology, advised by Prof. Xuechunzi Bai	
Fudan University	2022 — 2025
Master in Computer Science, advised by Prof. Yuan Zhang and Prof. Jiarun Dai	
Fudan University	2018 — 2022
B.S. in Department of Computer Science, majored in Information Security	

3. Publications

Published:

- **SCTrans: Constructing a Large Public Scenario Dataset for Simulation Testing of Autonomous Driving Systems**
Jiarun Dai*, **Bufan Gao***, Mingyuan Luo, Zongan Huang, Zhongrui Li, Yuan Zhang, Min Yang
In the 2024 International Conference on Software Engineering ([ICSE'24](#))
- **Measuring Bias or Measuring the Task: Understanding the Brittle Nature of LLM Gender Biases**
Bufan Gao, Elisa Kreiss
In the 2025 Conference on Empirical Methods in Natural Language Processing ([EMNLP'25](#))

In Preparation:

- **Collective Bias Emerges Even from Rational Social Learning**
Bufan Gao, Xuechunzi Bai
Under review
- **ExplorIt: Facilitating Simulation-based Fuzzing for Autonomous Driving Systems through Multi-Agent Interaction Modeling**
Bufan Gao, Zongan Huang, Jiarun Dai, Kairui Yang, Yixing Luo, Yingjie Fu, Yuan Zhang, Min Yang, Tao Xie
Under review of the Journal of ACM Transactions on Software Engineering and Methodology ([TOSEM](#))

4. Presentations

Collective Bias Emerges Even from Rational Social Learning, *spotlight data-blitz* in [SPSP 2026](#), Chicago
Microfoundations of Statistical Discrimination, *parallel talk* in [IC2S2 2025](#), Norrköping, Sweden
Measuring Bias or Measuring the Task: Understanding the Brittle Nature of LLM Gender Biases, *poster* in [EMNLP 2025](#), remote
SCTrans: Constructing a Large Public Scenario Dataset for Simulation Testing of Autonomous Driving Systems, main conference talk in [ICSE 2024](#), Lisbon, Portugal

5. Research Experiences

2024 - Present **Research Assistant, University of Chicago**
[Bai Lab](#) Working with Prof. [Xuechunzi Bai](#), I studies collective bias.

2024 - Present Research Assistant, UCLA

Coalas Lab While working with Prof. [Elisa Kreiss](#), I studied reliable measurements for bias in large language models (LLMs).

2021 - Present B.S. and Master Student, Fudan University

Secsys Lab Under the supervision of Prof. Min Yang, Prof. [Yuan Zhang](#) and Prof. [Jiarun Dai](#), with the focus on the *Safety and Robustness of Autonomous Driving Systems (ADS)*.

2021 Summer Computer Science Summer Research Fellowship Program, University of Chicago

Advised by Prof. Junchen Jiang and Prof. Nick Feamster

6. Honors & Achievements

Venustech Scholarship (Top 3% cross department), 2023

Champion of 2022 DEFCON AutoDriving Contest, 2022

Fudan University Freshman Scholarship, 2022

Champion of 2021 DEFCON AutoDriving Contest, 2021

Fudan University Liu Yong Ling Foundation Scholarship (Top 5% cross university), 2019

7. Service

ICSE 2025 workshop Committee Member

8. Teaching

Principles of Reverse Engineering (PoRE), Teaching Assistant

Spring 2023, Spring 2022